CORE PURE / PROOF BY INDUCTION

Proof by Induction - Derivatives

Worked solutions 8 questions · 54 marks

Questions

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1. It is given that

$$y = x^2 e^{ax}$$

where a is a constant.

Prove by mathematical induction that, for all integers $n \geq 2$,

$$\frac{\mathrm{d}^n y}{\mathrm{d}x^n} = a^{n-2}\left(a^2x^2 + 2nax + n(n-1)\right)e^{ax} \qquad \textbf{[6]}$$

Solution

Let

$$P(n): \quad \frac{\mathrm{d}^n y}{\mathrm{d}x^n} = a^{n-2}\big(a^2x^2 + 2nax + n(n-1)\big)e^{ax}$$

We prove $P(n)$ is true for all integers $n \geq 2$ by mathematical induction.

Base case: $n = 2$

Since $y = x^2e^{ax}$,

$$\begin{aligned}\frac{\mathrm{d}y}{\mathrm{d}x} &= 2xe^{ax} + x^2(ae^{ax}) \\ &= (ax^2 + 2x)e^{ax}\end{aligned}$$

Differentiate again:

$$\begin{aligned}\frac{\mathrm{d}^2y}{\mathrm{d}x^2} &= \frac{\mathrm{d}}{\mathrm{d}x}\big((ax^2 + 2x)e^{ax}\big) \\ &= (2ax + 2)e^{ax} + (ax^2 + 2x)(ae^{ax}) \\ &= (2ax + 2 + a^2x^2 + 2ax)e^{ax} \\ &= (a^2x^2 + 4ax + 2)e^{ax}\end{aligned}$$

Now substitute $n = 2$ into the given formula:

$$\begin{aligned}a^{2-2}\big(a^2x^2 + 2(2)ax + 2(2-1)\big)e^{ax} &= a^0(a^2x^2 + 4ax + 2)e^{ax} \\ &= (a^2x^2 + 4ax + 2)e^{ax}\end{aligned}$$

This matches $\frac{\mathrm{d}^2y}{\mathrm{d}x^2}$, so $P(2)$ is true.

Inductive Step

Now assume $P(k)$ is true for some integer $k \geq 2$. That is, assume

$$\frac{\mathrm{d}^k y}{\mathrm{d}x^k} = a^{k-2}\big(a^2x^2 + 2kax + k(k-1)\big)e^{ax}$$

This is the induction hypothesis.

We must show that $P(k+1)$ is true.

Differentiate the induction hypothesis:

$$\frac{\mathrm{d}^{k+1} y}{\mathrm{d}x^{k+1}} = a^{k-2}\frac{\mathrm{d}}{\mathrm{d}x}\big[\big(a^2x^2 + 2kax + k(k-1)\big)e^{ax}\big]$$

Using the product rule,

$$\begin{aligned}\frac{\mathrm{d}^{k+1}y}{\mathrm{d}x^{k+1}} &= a^{k-2}\big[\big(2a^2x+2ka\big)e^{ax}+\big(a^2x^2+2kax+k(k-1)\big)ae^{ax}\big]\\ &= a^{k-2}\big[2a^2x+2ka+a^3x^2+2ka^2x+ak(k-1)\big]e^{ax}\end{aligned}$$

Now simplify the bracket:

$$\begin{aligned}2a^2x+2ka+a^3x^2+2ka^2x+ak(k-1) &= a^3x^2+(2+2k)a^2x+\big(2k+k(k-1)\big)a\\ &= a^3x^2+2(k+1)a^2x+k(k+1)a\end{aligned}$$

So

$$\begin{aligned}\frac{\mathrm{d}^{k+1}y}{\mathrm{d}x^{k+1}} &= a^{k-2}\big(a^3x^2+2(k+1)a^2x+k(k+1)a\big)e^{ax}\\ &= a^{k-1}\big(a^2x^2+2(k+1)ax+k(k+1)\big)e^{ax}\end{aligned}$$

Since

$$k(k+1) = (k+1)\big((k+1)-1\big)$$

this becomes

$$\frac{\mathrm{d}^{k+1}y}{\mathrm{d}x^{k+1}} = a^{(k+1)-2}\big(a^2x^2+2(k+1)ax+(k+1)\big((k+1)-1\big)\big)e^{ax}$$

This is exactly $P(k+1)$, so the inductive step is complete.

Conclusion:

Therefore, since $P(2)$ is true and $P(k) \Rightarrow P(k+1)$, it follows by mathematical induction that, for all integers $n \geq 2$,

$$\frac{\mathrm{d}^{n}y}{\mathrm{d}x^{n}} = a^{n-2}\big(a^2x^2+2nax+n(n-1)\big)e^{ax}$$

2. Prove by mathematical induction that, for every positive integer n,

$$\frac{\mathrm{d}^n}{\mathrm{d}x^n}(x^3e^{-x}) = (-1)^n e^{-x}\big(x^3 - 3nx^2 + 3n(n-1)x - n(n-1)(n-2)\big) \qquad [7]$$

Solution

We prove the result by mathematical induction.

Let

$$P(n): \quad \frac{\mathrm{d}^n}{\mathrm{d}x^n}(x^3e^{-x}) = (-1)^n e^{-x}\big(x^3 - 3nx^2 + 3n(n-1)x - n(n-1)(n-2)\big)$$

We must show that $P(n)$ is true for every positive integer n.

Base case: $n = 1$**.**

Differentiate x^3e^{-x} using the product rule:

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}x}(x^3e^{-x}) &= 3x^2e^{-x} + x^3(-e^{-x}) \\ &= e^{-x}(3x^2 - x^3) \\ &= (-1)e^{-x}(x^3 - 3x^2)\end{aligned}$$

Now substitute $n = 1$ into the required formula:

$$(-1)^1e^{-x}\big(x^3 - 3(1)x^2 + 3(1)(0)x - (1)(0)(-1)\big) = (-1)e^{-x}(x^3 - 3x^2)$$

This matches the derivative found above, so $P(1)$ is true.

Inductive hypothesis.

Assume that $P(k)$ is true for some positive integer k.

That is, assume

$$\frac{\mathrm{d}^k}{\mathrm{d}x^k}(x^3e^{-x}) = (-1)^k e^{-x}\big(x^3 - 3kx^2 + 3k(k-1)x - k(k-1)(k-2)\big)$$

For convenience, let

$$P_k(x) = x^3 - 3kx^2 + 3k(k-1)x - k(k-1)(k-2)$$

Then the inductive hypothesis is

$$\frac{\mathrm{d}^k}{\mathrm{d}x^k}(x^3e^{-x}) = (-1)^k e^{-x}P_k(x)$$

Inductive step.

We now show that $P(k+1)$ is true.

Differentiate the inductive hypothesis:

$$\begin{aligned}\frac{\mathrm{d}^{k+1}}{\mathrm{d}x^{k+1}}(x^3e^{-x}) &= (-1)^k\frac{\mathrm{d}}{\mathrm{d}x}\big(e^{-x}P_k(x)\big) \\ &= (-1)^k\big((-e^{-x})P_k(x) + e^{-x}P_k'(x)\big) \\ &= (-1)^k e^{-x}\big(P_k'(x) - P_k(x)\big) \\ &= (-1)^{k+1}e^{-x}\big(P_k(x) - P_k'(x)\big)\end{aligned}$$

Now find $P_k'(x)$:

$$P_k'(x) = 3x^2 - 6kx + 3k(k-1)$$

So

$$\begin{aligned} P_k(x) - P_k'(x) &= \big(x^3 - 3kx^2 + 3k(k-1)x - k(k-1)(k-2)\big) \\ &\quad - \big(3x^2 - 6kx + 3k(k-1)\big) \\ &= x^3 - 3(k+1)x^2 + \big(3k(k-1) + 6k\big)x \\ &\quad - \big(k(k-1)(k-2) + 3k(k-1)\big) \\ &= x^3 - 3(k+1)x^2 + 3k(k+1)x - k(k-1)(k+1) \\ &= x^3 - 3(k+1)x^2 + 3(k+1)k\,x - (k+1)k\big((k+1) - 2\big) \end{aligned}$$

Therefore

$$\frac{\mathrm{d}^{k+1}}{\mathrm{d}x^{k+1}}(x^3e^{-x}) = (-1)^{k+1}e^{-x}\big(x^3 - 3(k+1)x^2 + 3(k+1)k\,x - (k+1)k((k+1) - 2)\big)$$

This is exactly the required formula with $n = k+1$, so $P(k+1)$ is true.

Conclusion.

Since $P(1)$ is true, and $P(k) \Rightarrow P(k+1)$, the statement holds for all positive integers n by mathematical induction.

Hence

$$\frac{\mathrm{d}^n}{\mathrm{d}x^n}(x^3e^{-x}) = (-1)^n e^{-x}\big(x^3 - 3nx^2 + 3n(n-1)x - n(n-1)(n-2)\big)$$

for every positive integer n.

3. Given that

$$y = e^{-2x}\cosh 3x$$

prove by induction that for $n \in \mathbb{N}$

$$\frac{\mathrm{d}^n y}{\mathrm{d}x^n} = e^{-2x}\left(\frac{1+(-5)^n}{2}\cosh 3x + \frac{1-(-5)^n}{2}\sinh 3x\right) \quad \textbf{[6]}$$

Solution

We prove the result by induction on n.

Let $P(n)$ be the statement

$$\frac{\mathrm{d}^n y}{\mathrm{d}x^n} = e^{-2x}\left(\frac{1+(-5)^n}{2}\cosh 3x + \frac{1-(-5)^n}{2}\sinh 3x\right)$$

where

$$y = e^{-2x}\cosh 3x$$

Base case: $n = 1$

Differentiate y using the product rule. Also,

$$\frac{\mathrm{d}}{\mathrm{d}x}\big(\cosh 3x\big) = 3\sinh 3x$$

So

$$\begin{aligned}\frac{\mathrm{d}y}{\mathrm{d}x} &= \frac{\mathrm{d}}{\mathrm{d}x}\big(e^{-2x}\cosh 3x\big)\\ &= e^{-2x}(-2)\cosh 3x + e^{-2x}(3\sinh 3x)\\ &= e^{-2x}\big(-2\cosh 3x + 3\sinh 3x\big)\end{aligned}$$

Now check this against the required formula when $n = 1$:

$$\frac{1+(-5)^1}{2} = \frac{1-5}{2} = -2, \qquad \frac{1-(-5)^1}{2} = \frac{1+5}{2} = 3$$

So

$$\frac{\mathrm{d}y}{\mathrm{d}x} = e^{-2x}\left(\frac{1+(-5)^1}{2}\cosh 3x + \frac{1-(-5)^1}{2}\sinh 3x\right)$$

Hence $P(1)$ is true.

Inductive hypothesis

Assume that $P(n)$ is true for some $n \in \mathbb{N}$. That is, assume

$$\frac{\mathrm{d}^n y}{\mathrm{d}x^n} = e^{-2x}\left(\frac{1+(-5)^n}{2}\cosh 3x + \frac{1-(-5)^n}{2}\sinh 3x\right)$$

For simplicity, write

$$A = \frac{1+(-5)^n}{2}, \qquad B = \frac{1-(-5)^n}{2}$$

Then the inductive hypothesis becomes

$$\frac{\mathrm{d}^n y}{\mathrm{d}x^n} = e^{-2x}(A\cosh 3x + B\sinh 3x)$$

Inductive step

We now differentiate to find $\frac{\mathrm{d}^{n+1}y}{\mathrm{d}x^{n+1}}$:

$$\begin{aligned}\frac{\mathrm{d}^{n+1}y}{\mathrm{d}x^{n+1}} &= \frac{\mathrm{d}}{\mathrm{d}x}\left[e^{-2x}(A\cosh 3x + B\sinh 3x)\right] \\ &= -2e^{-2x}(A\cosh 3x + B\sinh 3x) + e^{-2x}(3A\sinh 3x + 3B\cosh 3x) \\ &= e^{-2x}[(-2A+3B)\cosh 3x + (3A-2B)\sinh 3x]\end{aligned}$$

Now simplify the coefficients.

First,

$$\begin{aligned}-2A+3B &= -2\left(\frac{1+(-5)^n}{2}\right) + 3\left(\frac{1-(-5)^n}{2}\right) \\ &= \frac{-2-2(-5)^n+3-3(-5)^n}{2} \\ &= \frac{1-5(-5)^n}{2} \\ &= \frac{1+(-5)^{n+1}}{2}\end{aligned}$$

since

$$(-5)^{n+1} = -5(-5)^n$$

Also,

$$\begin{aligned}3A-2B &= 3\left(\frac{1+(-5)^n}{2}\right) - 2\left(\frac{1-(-5)^n}{2}\right) \\ &= \frac{3+3(-5)^n-2+2(-5)^n}{2} \\ &= \frac{1+5(-5)^n}{2} \\ &= \frac{1-(-5)^{n+1}}{2}\end{aligned}$$

Therefore

$$\frac{\mathrm{d}^{n+1}y}{\mathrm{d}x^{n+1}} = e^{-2x}\left(\frac{1+(-5)^{n+1}}{2}\cosh 3x + \frac{1-(-5)^{n+1}}{2}\sinh 3x\right)$$

So $P(n+1)$ is true.

Since $P(1)$ is true, and $P(n) \Rightarrow P(n+1)$, the result follows by induction for all $n \in \mathbb{N}$.

Hence

$$\frac{\mathrm{d}^n y}{\mathrm{d}x^n} = e^{-2x}\left(\frac{1+(-5)^n}{2}\cosh 3x + \frac{1-(-5)^n}{2}\sinh 3x\right) \quad \text{for all } n \in \mathbb{N}$$

4. Let

$$y = x^2 \cosh x$$

Prove by induction that, for all integers $n \geq 1$,

$$\frac{\mathrm{d}^{2n} y}{\mathrm{d}x^{2n}} = x^2 \cosh x + 4nx \sinh x + 2n(2n-1) \cosh x \qquad \textbf{[6]}$$

Solution

Let

$$P(n): \quad \frac{\mathrm{d}^{2n} y}{\mathrm{d}x^{2n}} = x^2 \cosh x + 4nx \sinh x + 2n(2n-1) \cosh x$$

We prove $P(n)$ by mathematical induction for all integers $n \geq 1$.

Base case: $n = 1$**.**

Since $y = x^2 \cosh x$,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = 2x \cosh x + x^2 \sinh x$$

and so

$$\begin{aligned} \frac{\mathrm{d}^2 y}{\mathrm{d}x^2} &= \frac{\mathrm{d}}{\mathrm{d}x}\left(2x \cosh x + x^2 \sinh x\right) \\ &= (2 \cosh x + 2x \sinh x) + (2x \sinh x + x^2 \cosh x) \\ &= x^2 \cosh x + 4x \sinh x + 2 \cosh x \end{aligned}$$

This is

$$x^2 \cosh x + 4(1)x \sinh x + 2 \cdot 1(2 \cdot 1 - 1) \cosh x$$

so $P(1)$ is true.

Inductive step.

Assume $P(k)$ is true for some integer $k \geq 1$. Then

$$\frac{\mathrm{d}^{2k} y}{\mathrm{d}x^{2k}} = x^2 \cosh x + 4kx \sinh x + 2k(2k-1) \cosh x$$

We must show that $P(k+1)$ is true.

Differentiate once:

$$\begin{aligned} \frac{\mathrm{d}^{2k+1} y}{\mathrm{d}x^{2k+1}} &= \frac{\mathrm{d}}{\mathrm{d}x}\left(x^2 \cosh x + 4kx \sinh x + 2k(2k-1) \cosh x\right) \\ &= (2x \cosh x + x^2 \sinh x) + 4k(\sinh x + x \cosh x) + 2k(2k-1) \sinh x \\ &= x^2 \sinh x + (4k+2)x \cosh x + \left(4k + 2k(2k-1)\right) \sinh x \\ &= x^2 \sinh x + (4k+2)x \cosh x + 2k(2k+1) \sinh x \end{aligned}$$

Differentiate again:

$$\begin{aligned}\frac{\mathrm{d}^{2k+2}y}{\mathrm{d}x^{2k+2}} &= \frac{\mathrm{d}}{\mathrm{d}x}\big(x^2\sinh x + (4k+2)x\cosh x + 2k(2k+1)\sinh x\big)\\ &= (2x\sinh x + x^2\cosh x) + (4k+2)(\cosh x + x\sinh x) + 2k(2k+1)\cosh x\\ &= x^2\cosh x + \big(2 + (4k+2)\big)x\sinh x + \big((4k+2) + 2k(2k+1)\big)\cosh x\\ &= x^2\cosh x + 4(k+1)x\sinh x + \big(4k+2+4k^2+2k\big)\cosh x\\ &= x^2\cosh x + 4(k+1)x\sinh x + (4k^2+6k+2)\cosh x\end{aligned}$$

Now

$$4k^2+6k+2 = 2(k+1)(2k+1) = 2(k+1)\big(2(k+1)-1\big)$$

So

$$\frac{\mathrm{d}^{2k+2}y}{\mathrm{d}x^{2k+2}} = x^2\cosh x + 4(k+1)x\sinh x + 2(k+1)\big(2(k+1)-1\big)\cosh x$$

This is exactly $P(k+1)$. Therefore $P(k) \Rightarrow P(k+1)$.

Since $P(1)$ is true, and $P(k) \Rightarrow P(k+1)$, it follows by induction that for all integers $n \geq 1$,

$$\frac{\mathrm{d}^{2n}y}{\mathrm{d}x^{2n}} = x^2\cosh x + 4nx\sinh x + 2n(2n-1)\cosh x$$

5. It is given that

$$y = (ax+1)^2 \ln(ax+1)$$

where a is a positive constant. Prove by mathematical induction that, for every integer $n \geq 3$,

$$\frac{\mathrm{d}^n y}{\mathrm{d}x^n} = 2(-1)^{n-3}\frac{(n-3)!a^n}{(ax+1)^{n-2}}$$ **[6]**

Solution

Let

$$P(n): \quad \frac{\mathrm{d}^n y}{\mathrm{d}x^n} = 2(-1)^{n-3}\frac{(n-3)!a^n}{(ax+1)^{n-2}}$$

We prove $P(n)$ is true for all integers $n \geq 3$ by mathematical induction.

Base case: $n = 3$**.**

Given

$$y = (ax+1)^2 \ln(ax+1)$$

Differentiate step by step.

$$\begin{aligned}\frac{\mathrm{d}y}{\mathrm{d}x} &= \frac{\mathrm{d}}{\mathrm{d}x}\big((ax+1)^2\big)\ln(ax+1) + (ax+1)^2\frac{\mathrm{d}}{\mathrm{d}x}\big(\ln(ax+1)\big)\\ &= 2(ax+1)\cdot a\cdot\ln(ax+1) + (ax+1)^2\cdot\frac{a}{ax+1}\\ &= 2a(ax+1)\ln(ax+1) + a(ax+1)\\ &= a(ax+1)\big(2\ln(ax+1)+1\big)\end{aligned}$$

Now differentiate again.

$$\begin{aligned}\frac{\mathrm{d}^2 y}{\mathrm{d}x^2} &= a\frac{\mathrm{d}}{\mathrm{d}x}\big[(ax+1)\big(2\ln(ax+1)+1\big)\big]\\ &= a\left[\frac{\mathrm{d}}{\mathrm{d}x}(ax+1)\big(2\ln(ax+1)+1\big) + (ax+1)\frac{\mathrm{d}}{\mathrm{d}x}\big(2\ln(ax+1)+1\big)\right]\\ &= a\left[a\big(2\ln(ax+1)+1\big) + (ax+1)\left(2\cdot\frac{a}{ax+1}\right)\right]\\ &= a\big[a\big(2\ln(ax+1)+1\big)+2a\big]\\ &= a^2\big(2\ln(ax+1)+3\big)\end{aligned}$$

Differentiate a third time.

$$\begin{aligned}\frac{\mathrm{d}^3 y}{\mathrm{d}x^3} &= a^2\frac{\mathrm{d}}{\mathrm{d}x}\big(2\ln(ax+1)+3\big)\\ &= a^2\left(2\cdot\frac{a}{ax+1}\right)\\ &= \frac{2a^3}{ax+1}\end{aligned}$$

Now substitute $n = 3$ into the claimed formula.

$$2(-1)^{3-3}\frac{(3-3)!a^3}{(ax+1)^{3-2}} = 2(-1)^0\frac{0!\,a^3}{(ax+1)^1}$$
$$= \frac{2a^3}{ax+1}$$

This agrees with $\frac{\mathrm{d}^3y}{\mathrm{d}x^3}$, so $P(3)$ is true.

Induction hypothesis.

Assume that for some integer $k \geq 3$,

$$\frac{\mathrm{d}^k y}{\mathrm{d}x^k} = 2(-1)^{k-3}\frac{(k-3)!a^k}{(ax+1)^{k-2}}$$

Inductive step.

We must show that this implies

$$\frac{\mathrm{d}^{k+1} y}{\mathrm{d}x^{k+1}} = 2(-1)^{k-2}\frac{(k-2)!a^{k+1}}{(ax+1)^{k-1}}$$

Starting from the induction hypothesis,

$$\frac{\mathrm{d}^{k+1}y}{\mathrm{d}x^{k+1}} = \frac{\mathrm{d}}{\mathrm{d}x}\Big(2(-1)^{k-3}(k-3)!a^k(ax+1)^{-(k-2)}\Big)$$
$$= 2(-1)^{k-3}(k-3)!a^k\frac{\mathrm{d}}{\mathrm{d}x}\Big((ax+1)^{-(k-2)}\Big)$$
$$= 2(-1)^{k-3}(k-3)!a^k\Big(-(k-2)(ax+1)^{-(k-1)}\cdot a\Big)$$
$$= 2\big[-(-1)^{k-3}\big](k-2)(k-3)!a^{k+1}(ax+1)^{-(k-1)}$$
$$= 2(-1)^{k-2}(k-2)!a^{k+1}(ax+1)^{-(k-1)}$$
$$= 2(-1)^{k-2}\frac{(k-2)!a^{k+1}}{(ax+1)^{k-1}}$$

This is exactly the required form for $P(k+1)$.

So, assuming $P(k)$ is true has allowed us to prove $P(k+1)$ is true.

Conclusion.

Since $P(3)$ is true, and $P(k) \Rightarrow P(k+1)$ for all integers $k \geq 3$, it follows by mathematical induction that

$$\frac{\mathrm{d}^n y}{\mathrm{d}x^n} = 2(-1)^{n-3}\frac{(n-3)!a^n}{(ax+1)^{n-2}} \qquad \text{for all integers } n \geq 3$$

Hence the result is proved.

6. A function is defined by $y = f(t)$ where $f(t) = (1+at)\ln(1+at)$ and a is a constant.

(a) By considering $\frac{\mathrm{d}^2y}{\mathrm{d}t^2}$, $\frac{\mathrm{d}^3y}{\mathrm{d}t^3}$, $\frac{\mathrm{d}^4y}{\mathrm{d}t^4}$ and $\frac{\mathrm{d}^5y}{\mathrm{d}t^5}$, make a conjecture for a general formula for $\frac{\mathrm{d}^ny}{\mathrm{d}t^n}$ in terms of n, a and t for any integer $n \geq 2$. **[3]**

(b) Use induction to prove the formula conjectured in part (a). **[4]**

(c) In the case when $f(t) = (1+3t)\ln(1+3t)$, find the rate at which the 6th derivative of $f(t)$ is varying when $t = \frac{5}{3}$. **[2]**

Solution

(a) Let

$$y = (1+at)\ln(1+at)$$

First differentiate once:

$$\begin{aligned}\frac{\mathrm{d}y}{\mathrm{d}t} &= \frac{\mathrm{d}}{\mathrm{d}t}(1+at)\cdot\ln(1+at) + (1+at)\cdot\frac{\mathrm{d}}{\mathrm{d}t}\big(\ln(1+at)\big)\\ &= a\ln(1+at) + (1+at)\cdot\frac{a}{1+at}\\ &= a\ln(1+at) + a\end{aligned}$$

Now find the next derivatives:

$$\frac{\mathrm{d}^2y}{\mathrm{d}t^2} = \frac{\mathrm{d}}{\mathrm{d}t}\big(a\ln(1+at)+a\big) = a\cdot\frac{a}{1+at} = \frac{a^2}{1+at}$$

$$\frac{\mathrm{d}^3y}{\mathrm{d}t^3} = \frac{\mathrm{d}}{\mathrm{d}t}\big(a^2(1+at)^{-1}\big) = a^2(-1)(1+at)^{-2}\cdot a = -\frac{a^3}{(1+at)^2}$$

$$\frac{\mathrm{d}^4y}{\mathrm{d}t^4} = \frac{\mathrm{d}}{\mathrm{d}t}\big(-a^3(1+at)^{-2}\big) = -a^3(-2)(1+at)^{-3}\cdot a = \frac{2a^4}{(1+at)^3}$$

$$\frac{\mathrm{d}^5y}{\mathrm{d}t^5} = \frac{\mathrm{d}}{\mathrm{d}t}\big(2a^4(1+at)^{-3}\big) = 2a^4(-3)(1+at)^{-4}\cdot a = -\frac{6a^5}{(1+at)^4}$$

The pattern is:

$$\frac{\mathrm{d}^2y}{\mathrm{d}t^2} = \frac{a^2}{(1+at)^1},\quad \frac{\mathrm{d}^3y}{\mathrm{d}t^3} = -\frac{a^3}{(1+at)^2},\quad \frac{\mathrm{d}^4y}{\mathrm{d}t^4} = \frac{2a^4}{(1+at)^3},\quad \frac{\mathrm{d}^5y}{\mathrm{d}t^5} = -\frac{6a^5}{(1+at)^4}$$

and the numbers $1, 1, 2, 6$ are $0!, 1!, 2!, 3!$.

So the conjecture is

$$f^{(n)}(t) = \frac{(-1)^n(n-2)!a^n}{(1+at)^{n-1}} \qquad \text{for all integers } n \geq 2$$

(b) We prove by induction that

$$f^{(n)}(t) = \frac{(-1)^n(n-2)!a^n}{(1+at)^{n-1}} \qquad (n \geq 2)$$

Base case: $n = 2$.

From part (a),

$$f''(t) = \frac{a^2}{1+at}$$

The formula gives

$$\frac{(-1)^2(2-2)!a^2}{(1+at)^{2-1}} = \frac{1 \cdot 0!\, a^2}{1+at} = \frac{a^2}{1+at}$$

so the result is true for $n = 2$.

Induction hypothesis.

Assume that for some integer $k \geq 2$,

$$f^{(k)}(t) = \frac{(-1)^k(k-2)!a^k}{(1+at)^{\,k-1}}$$

Induction step.

Differentiate both sides:

$$\begin{aligned} f^{(k+1)}(t) &= \frac{\mathrm{d}}{\mathrm{d}t}\Big[(-1)^k(k-2)!a^k(1+at)^{-(k-1)}\Big] \\ &= (-1)^k(k-2)!a^k \cdot \big(-(k-1)(1+at)^{-k} \cdot a\big) \\ &= (-1)^{k+1}(k-1)(k-2)!a^{k+1}(1+at)^{-k} \\ &= \frac{(-1)^{k+1}(k-1)!a^{k+1}}{(1+at)^k} \end{aligned}$$

This is exactly the required formula with $n = k+1$, since

$$(k+1)-2 = k-1 \qquad \text{and} \qquad (k+1)-1 = k$$

So, if the formula is true for $n = k$, it is also true for $n = k+1$.

Since it is true for $n = 2$, it follows by induction that

$$f^{(n)}(t) = \frac{(-1)^n(n-2)!a^n}{(1+at)^{\,n-1}} \qquad \text{for all integers } n \geq 2$$

(c) Here $a = 3$, so

$$f^{(n)}(t) = \frac{(-1)^n(n-2)!3^n}{(1+3t)^{\,n-1}}$$

The rate at which the 6th derivative is varying is the 7th derivative:

$$f^{(7)}(t) = \frac{(-1)^7(7-2)!3^7}{(1+3t)^6} = -\frac{5!\,3^7}{(1+3t)^6}$$

When $t = \frac{5}{3}$,

$$1+3t = 1+3\cdot\frac{5}{3} = 6$$

Hence

$$\begin{aligned} f^{(7)}\left(\frac{5}{3}\right) &= -\frac{120 \cdot 3^7}{6^6} \\ &= -\frac{120 \cdot 3^7}{(2 \cdot 3)^6} \\ &= -\frac{120 \cdot 3}{2^6} \\ &= -\frac{360}{64} = -\frac{45}{8} \end{aligned}$$

Therefore, the rate is

$$-\frac{45}{8}$$

7. The function $f(x)$ is defined by $f(x) = (1+x)^{-\frac{1}{2}}$, for $x > -1$.

(a) Prove by mathematical induction that the nth derivative of $f(x)$, $f^{(n)}(x)$, for all $n \geq 1$, is given by

$$f^{(n)}(x) = \frac{(-1)^n(2n)!}{2^{2n}n!(1+x)^{n+\frac{1}{2}}}$$

[4]

(b) Hence prove that

$$(1+x)^{-\frac{1}{2}} = 1 - \frac{x}{2} + \frac{3x^2}{8} - \frac{5x^3}{16} + \ldots + (-1)^n \frac{(2n)!}{2^{2n}(n!)^2}x^n + \ldots$$

for $-1 < x \leq 1$. **[3]**

Solution

(a) We prove the formula by mathematical induction on n.

Base case: $n = 1$

$$f(x) = (1+x)^{-1/2}$$

Differentiating once,

$$f'(x) = -\frac{1}{2}(1+x)^{-3/2}$$

The given formula gives, when $n = 1$,

$$\frac{(-1)^1(2)!}{2^2 \cdot 1!\,(1+x)^{1+\frac{1}{2}}} = \frac{-2}{4(1+x)^{3/2}} = -\frac{1}{2}(1+x)^{-3/2}$$

So the result is true for $n = 1$.

Induction hypothesis

Assume that for some $k \geq 1$,

$$f^{(k)}(x) = \frac{(-1)^k(2k)!}{2^{2k}k!(1+x)^{k+\frac{1}{2}}}$$

Induction step

Differentiate both sides:

$$\begin{aligned}
f^{(k+1)}(x) &= \frac{(-1)^k(2k)!}{2^{2k}k!} \cdot \frac{\mathrm{d}}{\mathrm{d}x}\left((1+x)^{-k-\frac{1}{2}}\right) \\
&= \frac{(-1)^k(2k)!}{2^{2k}k!}\left(-k - \frac{1}{2}\right)(1+x)^{-k-\frac{3}{2}} \\
&= \frac{(-1)^{k+1}(2k)!}{2^{2k}k!}\left(k + \frac{1}{2}\right)(1+x)^{-k-\frac{3}{2}} \\
&= \frac{(-1)^{k+1}(2k)!}{2^{2k}k!} \cdot \frac{2k+1}{2}(1+x)^{-k-\frac{3}{2}} \\
&= \frac{(-1)^{k+1}(2k)!(2k+1)}{2^{2k+1}k!}(1+x)^{-k-\frac{3}{2}}
\end{aligned}$$

Now simplify the coefficient:

$$\begin{aligned}\frac{(2k)!(2k+1)}{2^{2k+1}k!} &= \frac{2(k+1)(2k+1)(2k)!}{2^{2k+2}(k+1)k!} \\ &= \frac{(2k+2)(2k+1)(2k)!}{2^{2k+2}(k+1)!} \\ &= \frac{(2k+2)!}{2^{2k+2}(k+1)!}\end{aligned}$$

Hence

$$f^{(k+1)}(x) = \frac{(-1)^{k+1}(2k+2)!}{2^{2k+2}(k+1)!(1+x)^{k+\frac{3}{2}}}$$

which is exactly the required formula with $n = k+1$.

Therefore, by mathematical induction,

$$f^{(n)}(x) = \frac{(-1)^n(2n)!}{2^{2n}n!(1+x)^{n+\frac{1}{2}}} \qquad \text{for all } n \geq 1$$

(b) Using the Maclaurin series,

$$f(x) = f(0) + \sum_{n=1}^{\infty} \frac{f^{(n)}(0)}{n!}x^n$$

First,

$$f(0) = (1+0)^{-1/2} = 1$$

From part (a), for $n \geq 1$,

$$f^{(n)}(0) = \frac{(-1)^n(2n)!}{2^{2n}n!}$$

So

$$\begin{aligned}(1+x)^{-1/2} &= 1 + \sum_{n=1}^{\infty} \frac{1}{n!} \cdot \frac{(-1)^n(2n)!}{2^{2n}n!}x^n \\ &= 1 + \sum_{n=1}^{\infty} (-1)^n \frac{(2n)!}{2^{2n}(n!)^2}x^n\end{aligned}$$

Hence

$$(1+x)^{-1/2} = 1 - \frac{x}{2} + \frac{3x^2}{8} - \frac{5x^3}{16} + \cdots + (-1)^n \frac{(2n)!}{2^{2n}(n!)^2}x^n + \cdots$$

8. The function f is such that $f''(x) = -f(x)$.

Prove by mathematical induction that, for every positive integer n,

$$\frac{\mathrm{d}^{2n}}{\mathrm{d}x^{2n}}\left(x^2 f(x)\right) = (-1)^n\left(x^2 f(x) - 4nxf'(x) - 2n(2n-1)f(x)\right) \quad \textbf{[7]}$$

Solution

We prove the result by mathematical induction on n.

Let $P(n)$ be the statement

$$\frac{\mathrm{d}^{2n}}{\mathrm{d}x^{2n}}\left(x^2 f(x)\right) = (-1)^n\left(x^2 f(x) - 4nxf'(x) - 2n(2n-1)f(x)\right)$$

Base case: $n = 1$

We calculate the second derivative directly:

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}x}\left(x^2 f(x)\right) &= 2xf(x) + x^2 f'(x)\\ \frac{\mathrm{d}^2}{\mathrm{d}x^2}\left(x^2 f(x)\right) &= \frac{\mathrm{d}}{\mathrm{d}x}\left(2xf + x^2 f'\right)\\ &= 2f + 2xf' + 2xf' + x^2 f''\\ &= 2f + 4xf' + x^2 f''\end{aligned}$$

Given that $f''(x) = -f(x)$, this becomes

$$\begin{aligned}\frac{\mathrm{d}^2}{\mathrm{d}x^2}\left(x^2 f(x)\right) &= 2f + 4xf' - x^2 f\\ &= -(x^2 f - 4xf' - 2f)\\ &= (-1)^1\left(x^2 f - 4(1)xf' - 2(1)(2\cdot 1 - 1)f\right)\end{aligned}$$

So $P(1)$ is true.

Inductive hypothesis

Assume that $P(k)$ is true for some positive integer k. Then

$$\frac{\mathrm{d}^{2k}}{\mathrm{d}x^{2k}}\left(x^2 f\right) = (-1)^k\left(x^2 f - 4kxf' - 2k(2k-1)f\right)$$

Inductive step

We must show that $P(k+1)$ is true.

Starting from the inductive hypothesis and differentiating twice,

$$\begin{aligned}\frac{\mathrm{d}^{2k+2}}{\mathrm{d}x^{2k+2}}\left(x^2 f\right) &= \frac{\mathrm{d}^2}{\mathrm{d}x^2}\left[\frac{\mathrm{d}^{2k}}{\mathrm{d}x^{2k}}\left(x^2 f\right)\right]\\ &= (-1)^k\frac{\mathrm{d}^2}{\mathrm{d}x^2}\left(x^2 f - 4kxf' - 2k(2k-1)f\right)\\ &= (-1)^k\left((x^2 f)'' - 4k(xf')'' - 2k(2k-1)f''\right)\end{aligned}$$

Now work out each of these second derivatives.

First, from the base case calculation,

$$(x^2 f)'' = -x^2 f + 4xf' + 2f$$

Next,

$$\begin{aligned}(xf')' &= f' + xf'' \\ (xf')'' &= f'' + f'' + xf''' \\ &= 2f'' + xf'''\end{aligned}$$

Since $f'' = -f$, differentiating gives $f''' = -f'$. Therefore

$$(xf')'' = 2(-f) + x(-f') = -2f - xf'$$

so

$$(xf')'' = -xf' - 2f$$

Also,

$$f'' = -f$$

Substituting these into the inductive step,

$$\begin{aligned}\frac{\mathrm{d}^{2k+2}}{\mathrm{d}x^{2k+2}}\left(x^2 f\right) &= (-1)^k\left[(-x^2 f + 4xf' + 2f) - 4k(-xf' - 2f) - 2k(2k-1)(-f)\right] \\ &= (-1)^k\left[-x^2 f + 4xf' + 2f + 4kxf' + 8kf + 2k(2k-1)f\right] \\ &= (-1)^k\left[-x^2 f + 4(k+1)xf' + \left(2 + 8k + 2k(2k-1)\right)f\right]\end{aligned}$$

Now simplify the coefficient of f:

$$\begin{aligned}2 + 8k + 2k(2k-1) &= 2 + 8k + 4k^2 - 2k \\ &= 4k^2 + 6k + 2 \\ &= 2(k+1)(2k+1)\end{aligned}$$

Hence

$$\begin{aligned}\frac{\mathrm{d}^{2k+2}}{\mathrm{d}x^{2k+2}}\left(x^2 f\right) &= (-1)^k\left[-x^2 f + 4(k+1)xf' + 2(k+1)(2k+1)f\right] \\ &= (-1)^{k+1}\left[x^2 f - 4(k+1)xf' - 2(k+1)(2k+1)f\right]\end{aligned}$$

Since $2(k+1) - 1 = 2k + 1$, this is exactly

$$\frac{\mathrm{d}^{2(k+1)}}{\mathrm{d}x^{2(k+1)}}\left(x^2 f\right) = (-1)^{k+1}\left(x^2 f - 4(k+1)xf' - 2(k+1)(2(k+1)-1)f\right)$$

So $P(k+1)$ is true.

Conclusion

$P(1)$ is true, and $P(k) \Rightarrow P(k+1)$. Therefore, by mathematical induction,

$$\frac{\mathrm{d}^{2n}}{\mathrm{d}x^{2n}}\left(x^2 f(x)\right) = (-1)^n\left(x^2 f(x) - 4nxf'(x) - 2n(2n-1)f(x)\right)$$

for every positive integer n.